Generating-function solutions

DM09-E01

Differentiate the geometric series and multiply by x:

$$x/(1-x)^2 = \sum_{n\geq 1} nx^n.$$

The constant is zero, as $a_0=0$. Multiplication by x aligns the derivative's shifted indices with the original sequence.

DM09-E02

A(x)=(1−x)$^{-2}$ and B(x)=(1−x)$^{-1}$. Their product is $(1-x)^{-3} = \sum_{n\geq 0} \binom{n+2}{2} x^n$, giving $\binom{6}{2} = 15$ at n=4. This is also 1+2+3+4+5=15, the prefix sum of a_n.

DM09-E03

Choosing value j for coordinate i contributes weight ij, hence factor $\sum_{j\geq 0} x^{ij} = 1/(1-x^i)$. Here x is an indeterminate, distinct from the solution coordinates. Multiplying the factors for i=1,2,3 records every triple once, with degree $j_1+2j_2+3j_3$. The degree-n coefficient therefore counts exactly the required solutions.

DM09-E04

Write parts in order a≥b≥0, interpreting zero as an absent part. Set d=a−b≥0; then n=2b+d. This bijects with choices of b twos and d ones, giving OGF 1/((1−x)(1−x^2)) and coefficient ⌊n/2⌋+1. At n=4 the partitions are 4+0,3+1,2+2. The convention includes the empty partition at zero.

DM09-E05

Each optional singleton type has EGF 1+x; their labeled product is (1+x)^2=1+2x+x^2. Its ordinary x^2 coefficient is one, but two specified labels form 2! $[x^2](1+x)^2 = 2$ structures by swapping their assignment to types. At size one there are two type choices. Choosing a subset of two fixed named objects happens to give the same ordinary polynomial, but has a different model and no multiplication by n!.

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DM09-E06

The prefix OGF is $P(x) = A(x)/(1-x) = \sum_{n\ge 0}(\sum_{k=0}^{n} a_k)x^n$. Therefore

$$xP'(x) = \sum_{n\ge 1} n\left(\sum_{k=0}^{n} a_k\right)x^n,$$

so the desired OGF is $x\frac{d}{dx}(A(x)/(1-x))$. Its degree-zero coefficient is automatically zero, consistent with t_0.

DM09-E07

For n<2 the coefficient is zero. For n≥2, shift to degree n−2 and use the negative binomial expansion:

$$[x^n]\frac{x^2}{(1-3x)^3} = [x^{n-2}](1-3x)^{-3} = 3^{n-2}\binom{n}{2}.$$

At n=2 this is one, the coefficient of the lowest term x^2. At n=3 it is $3\binom{3}{2} = 9$, matching the linear coefficient in the denominator's inverse cube.

DM09-E08

Work in $\mathbb{Q}[[x]]$, where the constant term 2 is invertible and the inverse exists uniquely. Over $\mathbb{Z}[[x]]$, 2 is not a unit, so no inverse exists. Require $(2-3x+x^2)(b_0+b_1x+b_2x^2+b_3x^3) \equiv 1 \pmod{x^4}$. Successive equations give

$$2b_0 = 1, \quad 2b_1 - 3b_0 = 0, \quad 2b_2 - 3b_1 + b_0 = 0, \quad 2b_3 - 3b_2 + b_1 = 0.$$

Thus b_0=1/2,b_1=3/4,b_2=7/8,b_3=15/16 and B(x)=1/2+3x/4+7x^2/8+15x^3/16+…. Multiplication through degree three gives

$$(1-3x/2+x^2/2)+(3x/2-9x^2/4+3x^3/4)+(7x^2/4-21x^3/8)+15x^3/8 = 1+0x+0x^2+0x^3.$$

No coefficients above degree three were requested.

DM09-E09

E(x)=Σx^r/r!, r≥1, counts one nonempty labeled block. Three factors initially distinguish three slots; division by 3! removes their order. On four labels, three nonempty unlabeled blocks must have sizes 2,1,1. Choose the pair in $\binom{4}{2} = 6$ ways, determining the rest. Hence

$$4!\,[x^4](e^x - 1)^3/3! = 6 = S(4,3).$$

The ordinary coefficient is 6/4!=1/4. The 3! removes block-slot order; 4! converts normalized EGF coefficients into counts on the four labels.

DM09-E10

Odd parts give $\prod_{m\ge1}(1 - x^{2m-1})^{-1}$, while distinct parts give $\prod_{k\ge1}(1 + x^k)$. Rewriting each latter factor as $(1-x^{2k})/(1-x^k)$ cancels all even denominator factors, leaving only the odd ones. Each degree depends on finitely many factors, making this formal cancellation valid.

At seven the odd partitions are 7,5+1+1,3+3+1,3+1^4,1^7. Distinct partitions are 7,6+1,5+2,4+3,4+2+1. Both lists have five members.

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DM09-E11

With nonnegative slack u_4, write $u_1+2u_2+3u_3+u_4=n$. The first and fourth variables give $1/(1-x)$, the others $1/(1-x^2)$,$1/(1-x^3)$, so

$$\sum_{n\ge 0} a_n x^n = 1/((1-x)^2(1-x^2)(1-x^3)).$$

Use $(1-x)^{-2}=\sum_{m\ge 0}(m+1)x^m$, $(1-x^2)^{-1}=1+x^2+x^4+x^6+\cdots$, and $(1-x^3)^{-1}=1+x^3+x^6+\cdots$.

j_2	j_3	$m=7-2j_2-3j_3$	Contribution $m+1$
0	0	7	8
0	1	4	5
0	2	1	2
1	0	5	6
1	1	2	3
2	0	3	4
2	1	0	1
3	0	1	2

The total $8+5+2+6+3+4+1+2=31$ gives a_7. Direct enumeration of triples with cost at most seven has the same rows: for each j_2,j_3, u_1 ranges from zero through m, giving $m+1$ choices. A three-factor denominator counts exact cost n only. The additional slack factor sums all remaining unused amounts, as required by at most n.

DM09-E12

Fixed k ordered component slots have EGF $\hat{A}(x)^k$. Disjoint nonempty label sets make all components distinct, so each unordered structure has exactly $k!$ slot orders. Divide by $k!$. With $\hat{A}(0)=0$, each component needs a label and degree n uses only $k\le n$, permitting the formal sum $\exp(\hat{A})$. Empty components would allow arbitrarily many empty blocks and undermine both finite layering and the free $k!$ ordering argument.

On four labels, ordered sizes $(1,3)$,$(2,2)$,$(3,1)$ contribute $4,6,4$. Total fourteen ordered structures divided by two gives seven unordered partitions. The ordinary x^4 coefficient is $7/4!=7/24$; the normalized structure count is seven.

DM09-E13

Multiplicities 3=1+2,2=2,5=1+4 replace three fives by 5+10, two threes by 6, five ones by 1+4, giving 10+6+5+4+1=26 with distinct parts. Inverse factorizations 10=2 • 5,6=2 • 3,5=1 • 5,4=4 • 1,1=1 • 1 restore multiplicities three fives, two threes, five ones.

For λ=(4,2,1), columns have lengths 3,2,1,1, so λ′=(3,2,1,1), still totaling seven. Its columns have lengths 4,2,1. Generally cell (i,j) exists iff $j \le \lambda_i$ iff $i \le \lambda'_j$. Swapping coordinates twice restores all cells, proving involution and exchanging largest part with number of parts.

DM09-E14

Letters specify subsets of positions; identical letters are not separately labeled. The EGF is $F(x)=(1+x+x^2/2)^2(1+x)$. The square is $1+2x+2x^2+x^3+x^4/4$; multiplying by 1+x gives x^3 coefficient three and x^4 coefficient 5/4. Counts are 3! • 3=18 and 4! • 5/4=30.

At length three, (2,1,0),(1,2,0),(2,0,1),(0,2,1) each give 3!/(2!1!)=3 and (1,1,1) gives six, totaling eighteen. At length four, (2,2,0) gives six and (2,1,1),(1,2,1) each give twelve, totaling thirty.

For even A-count at length four, cases zero,two,four give 16, $\binom{4}{2}2^2 = 24$,1, totaling 41. The EGF is $\cosh(x)e^{2x}=(e^{3x}+e^x)/2$, giving $(3^4+1)/2=41$ after multiplying its degree-four coefficient by 4!. Here cosh simply denotes the even-degree exponential series.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 9.

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